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IBRAHIM SOHIL SADI SAFI

Embedded Systems Engineer
AI & Computer Vision Specialist

CONTACT

LOCATION

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GITHUB

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TECHNICAL SKILLS

Programming Languages

C · C++ · Python · Java · JavaScript ·
Assembly · SQL

Embedded Systems

Microcontroller & microprocessor
programming · Firmware development ·
Circuit design · Electronics troubleshooting ·
Wired & wireless communication networks

Industrial Automation

PLC programming (Ladder Logic) · GRAFCET
design & simulation

AI & Data

Deep Learning · Computer Vision (NVIDIA
certified) · Data management · Statistics &
analysis

Tools & IT

Windows, Linux & macOS · MS Office · Visual
Basic · Cloud-based solutions

LANGUAGES

Arabic — Native

English — Professional working proficiency

French — Good working proficiency

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Electronics · Firmware · Industrial Automation · Deep Learning

PROFILE SUMMARY

Embedded Systems Engineer with a Master's degree in Electronics, combining strong low-level programming skills (C, C++, Assembly) with modern AI and computer-vision expertise certified by NVIDIA. Experienced in designing, building and troubleshooting microcontroller-based systems and industrial automation solutions (PLC, Ladder Logic, GRAFCET). Recognized for rigorous engineering, clear documentation and the ability to deliver under pressure, both independently and in team environments.

EXPERIENCE & PROJECTS

Real-Time Facial Expression Recognition System

2022 - 2023

Master's Thesis Project — Deep Learning & Artificial Vision

- ▶ Designed, trained and validated a deep-learning model in Python for real-time facial emotion recognition, achieving excellent accuracy.
- ▶ Deployed and evaluated the system in real-world scenarios; produced full technical documentation and defence materials.

Industrial Trainee — Electrical Networks

2021

SONELGAZ — National Electricity & Gas Company, Algeria (2-month intensive training)

- ▶ Hands-on training on protecting and connecting electrical networks to the highest safety standards.
- ▶ Worked with key electronic and electro-electrical components for residential and high-voltage installations.

Algae Cultivation for Clean Energy

2019

Mini-Grant Research Project

- ▶ Investigated algae farming as a sustainable feedstock for future clean-energy production.

"Talking Container" — Environmental Project

2014

Community Awareness Initiative

- ▶ Built an interactive "talking" waste container promoting environmental awareness and recycling.

EDUCATION

Master's Degree in Electronics — Embedded Systems

2021 - 2023

Bachelor's Degree in Electronics

2018 - 2021

Continuing Professional Development

2023 - Present

Web development, software engineering and factory automation with AI & advanced machine learning

CERTIFICATIONS & TRAINING

- ▶ Artificial Vision (Python) & Deep Learning — NVIDIA **2022**
- ▶ Introduction to Business Management — King's College London (Online) **2020**
- ▶ Programming Course: Java, C, C++ **2017**
- ▶ Language Studies: English & French — Royal School **2018**
- ▶ MS Office & IT Training (Word, Excel, Visual Basic) **2014**
- ▶ First Aid & Diabetes Care — Red Crescent / Red Cross **2012**

VOLUNTEERING

- ▶ **Volunteer — Palestinian Red Crescent Society (2014 - 2016):** emergency support and field coordination; performed effectively under high-pressure, critical situations.